Lesson Aim

Specify the techniques used in the culture of grape vines (Pruning and Training Grapevines).

Pruning And Training Vines

A major factor in the success of viticulture lies in the correct pruning and training of the vine. Pruning curbs the excessive growth of the vine and thus enables it to produce sufficient new fruitful shoots each year. Training ensures the vine grows in a manner that allows easy maintenance and harvest, and improves the quality of grape yields.

Training Vines

Grape vines do not have a thick trunk to support the plant above ground and, hence, require manmade support. Different training systems (or trellises) are commonly used to give the vines a support base. Tendrils growing from the main canes will wrap around provided support wires, allowing the vine to stay upright. Once supported, the vine can be trained for the purpose of maximising desirable growth characteristics. Trellises allow foliage maximum exposure to the sun and good airflow, and maintain the fruit and vines at a workable height.

Vine Pruning

The vigorous root system of a vine has the capacity to provide plentiful water and nutrients to the shoot. Excessive vegetative growth will often be the result. In order to maximise fruit yield and quality, vines need to be pruned to remove old canes and contro bud numbers.

Pruning is definitely beneficial and usually necessary for controlling the quality and quantity of grapes produced. The way grapes are pruned, however, can vary greatly from place to place around the world, depending upon what the grower hopes to achieve from the crop, as well as the conditions found locally.

Vines on some of the Greek Islands are pruned back hard to little more than a short, low stub. In other places a single vine can become a huge plant sprawling over a large trellis high in the air above head height. If mechanical harvesting is to be used, the vines need to be pruned differently again, to accommodate the machinery.

Generally about 90% of the previous season’s growth is removed in winter each year. The next season’s fruit will occur on the 10% of first-year wood left remaining on the vine.

Despite a reduction in the total plant size after pruning, the individual shoots, leaves and bunches of grapes which develop become larger.

Pruning decreases the number of bunches, but improves the size and quality of the individual grapes.

Regular, good pruning will greatly help a vine to be reliable in its production. A vine which is badly pruned is likely to crop very well one year and poorly the next. A vine which is pruned well is more likely to produce consistently good crops each year.

If a vine is lightly pruned it will produce a large crop of lower quality fruit that year but, more often than not, the next year's crop will suffer. Extra heavy pruning, on the other hand, can result in a lower crop yield, increased bunch size, and excessive growth of vegetative shoots. The aim is to prune to a level between these two extremes.

Vines are pruned for five reasons:

- To produce consistent cropping from one year to the next.
- To improve the quality of fruit.
- To create and maintain a desirable vine shape.
- To keep the vine growth young and vigorous
- To facilitate cultural practices such as spraying, irrigation, cultivation and harvesting.

The methods of pruning and training used will depend on the variety of grape and how and where the vine is to be grown.

There are two principle pruning techniques for vines: spur-pruning (also called cordon pruning) and rod-pruning (also called cane pruning).

For glasshouse culture of vines, only spur-pruning is used. Each year after harvest, all fruiting laterals (known as rods or canes) growing from the main laterals are cut back to two buds, leaving a knobby stub or spur. Two new fruiting laterals will grow from each of these spurs the following season.

Rod-pruning differs in that the main laterals and the fruiting laterals are cut back each year. Any main laterals that grew fruiting laterals during the first fruiting season are cut back to two buds after harvest; in two years' time replacement laterals will grow from those two buds. In the intervening period, two of the new laterals that developed during the first fruiting season are left to grow int the main laterals for the second fruiting season.

The simpler of the two methods, spur-pruning, is used on most, but not all, European grapes. Fruit production on fruiting laterals grown close to the main laterals is not reliable in some American cultivars and French hybrids. The main laterals must be renewe annually, by rod-pruning, on these vines.

Shoot Spacing

Shoots are pruned to promote even spacing. Evenly-spaced shoots allow the vine adequate exposure to sunlight and promote air circulation through the canes, leaves and fruit bunches. The benefit of good air circulation is the lowering of micro fluctuations in humidity. Trapped pockets of air among dense vegetation and over-clustered fruit can become saturated with moisture. Evaporating dew and mist may exacerbate this effect. Disease-causing organisms such as bacteria and fungi grow better in humi conditions. Under-pruned vines will often suffer from higher rates of foliar disease.

Adequately spaced shoots will not only develop less disease but also be easier to treat with sprays. Natural or synthetic pesticide sprays have a better opportunity to penetrate sparsely vegetated vines and give effective disease control.

Vine Spacing

Spacing of the vines along the row will also impact performance. Strong varieties grown in areas that are conducive to good growth will benefit from more generous spacing, eg. 2 metres (6 feet) between plants. If vigorous vines are not spaced adequately, excessive shading of leaves and fruit occurs, maintenance and harvesting become more difficult, and disease incidence will increase. Weaker-performing varieties, however, can be spaced as close as one metre (3 feet) apart along the row with no ill effect.

Bud Numbers

The level of pruning will determine the concentration of buds along the vine. Bud number has a direct impact not only on total yield but, also, on grape quality and flavour characteristics. Some growers will specify 'buds per hectare'; however, it is common to stipulate buds per metre (yard) of vine. High-yielding grape varieties, such as Riesling and Pinot Gris are pruned to 30-35 buds/metre (yard). Lower yielding varieties, such as Chardonnay or Merlot will perform better with as many as 50 buds/metre (yard). Bud numbers can be manipulated by either altering cane numbers or spur length. The specific level of pruning is best determined by the grower's observations of crop performance from previous years. A decrease in bud number would be justified if previous crops saw poor cane vigour, insufficient fruit ripeness, or poor intensity in flavour of the wine.

Prune vines to achieve the following:

- Space the shoots evenly to achieve good air circulation, good even sunlight to all parts of the plant, and hence, minimal humidity throughout the coming growing season.
- Space the shoots so that the sprays will penetrate the plant evenly.
- Make sure shoots which will regrow are located where they will be preferred, for replacement shoots to give the following season's crop.
- Choose the length and position of shoot where the bud has the best potential for fruiting.
- Aim to get the appropriate number of buds per plant to give the maximum yield.

How Much To Prune?

The fruit yield is partly affected by the number of buds left per hectare. This is normally expressed as buds per metre (or yard) of trellised plants (see following table).

Examples of Recommended Number of Buds per Metre of Trellis

Type of Grape	Buds/metre (yard)
High yielding e.g. Chenin Blanc	20-30
Moderate yielding e.g. Riesling	30-35
Low yielding e.g. Cabernet Sauvignon	35-45
Very low yielding e.g. Chardonnay	40-50

The above bud numbers are appropriate under "normal" conditions.

Decrease the number of buds left on vines when:

- Growth is consistently poorer than expected
- Grapes on those vines are not normally ripening as well as you hoped
- Harvest dates are usually later than what you had expected on the plantings
- Wine produced from the grapes is frequently poor quality

You should increase the number of buds left on a vine when:

- Growth is above average (due to ideal soil, nutrition, climate, water etc.)
- Rows are spaced wider than normal
- You are growing for quantity of harvest instead of quality (e.g. to produce bulk cheap wine)
- You are growing a low yield clone of a variety (e.g. some varieties of Riesling bear much higher than others)

Vines are pruned heavily leaving only a selection of the stronger new season canes.

Machine Pruning

Using machinery to cut vines will save greatly on the expense of pruning, but at a sacrifice: it is impossible to finely control where cuts are made. There are many in the industry who support mechanical pruning, and neither yield nor fruit quantity tend to suffer significantly. Repeated mechanical pruning can, however, cause the cordon to become congested with too many old stems, and in order to clean this up, a hand prune will become essential.

Summer Pruning

Pruning may be done once or several times after shoots have developed some significant bulk:

- Trimming just before flowers are ready to pollinate may be done to increase fruit set.
- Once shoots grow above the highest wire on a trellis, they may have the top cut out.
- If growth becomes too dense, some shoots may be partly or fully removed to allow better ventilation.

Combination Pruning

Some vineyards choose to hand prune bare vines over winter, but use mechanical cutters to do any pruning of plants in leaf.

Note: Though grapes have traditionally been pruned relatively hard, some research has indicated that less pruning may in fact provide advantages over heavy pruning. As with most things, don't necessarily assume that "because it has always been done that way" that other options are not appropriate in certain circumstances. The best viticulturist will always be a person with an open, inquiring and innovative approach to all that they do.

Pruning Sultana Vines

When pruning a sultana vine, consider the following:

- Consider the number of canes and quality of canes to be left on the vine.
- Consider the positioning of the canes to be left.
- Consider leaving spurs (i.e. the base of a removed cane) in preferred positions for replacement canes to grow from for the next year.
- The amount of material to be removed depends on the vigour of each plant. (Modify this according to the fruitfulness of each bud, which can vary from area to area, and year to year.)
- In years when the fruitfulness of buds is low more canes are left, but when the fruitfulness is high one or two canes fewer than is normal can be left. The average bud fruitfulness is 50% (i.e. 50% of buds on the canes have fruit - this can be seen by looking at dissected buds under a microscope).
- With an average bud count of 50% and good vigour, an established sultana vine would have 8-10 canes left on it, with each cane bearing about 14 buds (i.e. 112 to 140 buds per plant). In some situations longer canes are used.
- Bud positions 1 to 4 are not normally very fruitful (i.e. the buds closest to the trunk). These tend to produce mainly canes for the next season.
- Buds in positions 5 to 11 tend to be the most fruitful.
- The best fruiting canes are usually those which have a light brown matured colour and where the buds are close together (i.e. 8-12cm apart).

Stages

There are three stages to pruning a sultana vine:

Cutting out

This is where most of the canes to be removed are cut out. Always leave more than you want though, as some canes may break, some may be damaged and you will need to cut a few to leave spurs for the next season's growth. Always leave a stub of wood of each old cane to allow for die back to occur without spreading into the structure of the plant.

Pulling Out and Trimming Up

All the wood cut out by the pruner is pulled out, or removed. Any remaining canes can then be trimmed to remove laterals and tendrils, and if necessary have the tips cut to bring them to the preferred length. Remember that it is the total number of buds per vine which is left that is important.

Wrapping Down

This involves tying the canes into position on the trellis. Start with the cane nearest to the centre of the vine, and bring it to the wir along the most direct route. Don't plait canes around the wire. Wrap canes all in one direction around the wire(s). Don't cross any canes over the top of the crown (i.e. the centre of the vine), as this can induce disease, and it can reduce development of new growth close to the trunk.

Trellising Sultana Vines

Sultanas have been grown many different ways on different types of trellis systems, in different parts of the world. One common method in Australia involves a T-shaped trellis (in cross section), about 1 metre (3 feet) tall.

Vertical posts have horizontal timbers on top (making the T shape), and wires run between the posts attached to the horizontal top of the support. The crown of the vine (i.e. top of the trunk) is about 30cm (12 in) below the top of the trellis.

Trellising

Trellises are used to support the vine, allow foliage maximum exposure to the sun and air, and to maintain the fruit and vines at a workable height. The design of the trellis system depends on the crop, harvest techniques, expected yield of the crop and personal (or district) preference in the design.

Although there are a variety of systems available, vertical trellises are the most common. Support posts should generally not be higher than about 2 metres (6½ft) to enable easy pruning and harvesting. Maximising leaf exposure to sunlight is an important goal of vine training. North-south orientation of rows assists this.

In the first year of a vineyard, generally no trellis system is required. The training system should, however, be installed at planting time ready for use. For simple domestic training systems, during the first, second, third and possibly fourth year, all that may be required is a single wire approximately 1m (39in) above the ground, supported at each end by post systems and along the row by intermediate posts every 7-11 metres (23-36ft). As the vine grows, the wire system can be built up.

As the vine reaches fourth or fifth year (depending on the vigour of the vine), a second wire is added 20-25cm (8-10in) above the first wire and on the up-wind side of the post. This will act as the "foliage" wire.

In a few years another "foliage" wire is added above the previous wires. Instead of just one wire it is possible to use two wires on either side of the post to provide extra support for the leaves.

In high rainfall areas or where irrigation is used, the "T" structure is frequently used.

> You can train vines to grow in any shape you wish, but for ease of handling and to optimise commercial yields a standard system tends to be adopted within any one region. Look at what other growers do in your region, and this is usually a good place to start.

Other training systems are used throughout the world including the Guyot system, the Geneva Double Curtain (GDC), the Kniffen systems – four arm, six-arm and umbrella Kniffen systems, the single post system and the pergola system (see section following on Training Systems).

Examples of Different Ways of Training Grapes

Most Commercial vines are grown on trellis systems similar to this.

In some harsher climates, vines may be trained to grow lower on a trellis such as this, to minimise the effect of strong winds.

Another way of training a vine (growing on a single pole).

Vines being trained to cover an arch.

Container grown vine being trained on a wire frame.

Materials for Construction

End Assembly

Common materials used are: local timber posts; old tram or railway lines; steam or water pipes; old metal railway sleepers. The material must be strong enough to support the weight of the vines and affordable. The dimensions of posts vary according to the training systems used.

Three end assembly systems used are:

- Post and Stay End - probably the best, doesn't move much if constructed correctly, lasts about 40 years.
- Post and Rail End - solid engineering but not as strong as the previous system, especially if hit by a tractor.
- Post and Anchor System - is subject to damage and is a nuisance during cultivation as the anchor is in the headland and no protected in the row. The anchor can move in very wet conditions.

Intermediate Posts

The most popular material used for an intermediate post is treated pine 2.4m (7½ft) high and 8-10cm (3-5in) in diameter, although these dimensions vary with the training system employed. Cut hardwood saplings and sawn fence posts can also be used. Sharpened pine posts can be driven into soft soil with a tractor-mounted post driver if available, or one can be hired.

Wire

Galvanised wire is best used in the vineyard either low or high tensile grades.

Low tensile - soft, easy to work, but needs to be of thicker weight to do the same job, expensive, sags

High tensile - lighter, cheaper, difficult to work with, more likely to break, does not sag, provides a tighter neat trellis wire.

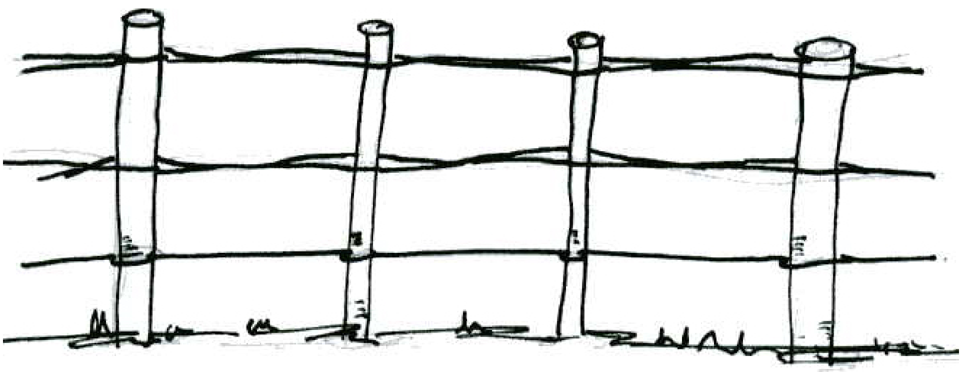
Attaching Wire to Posts

Wires are both wrapped around the end post, and tied, with the strainer on the wire close to the post, or they are passed through a hole and the wire strainer is placed on the back of the post.

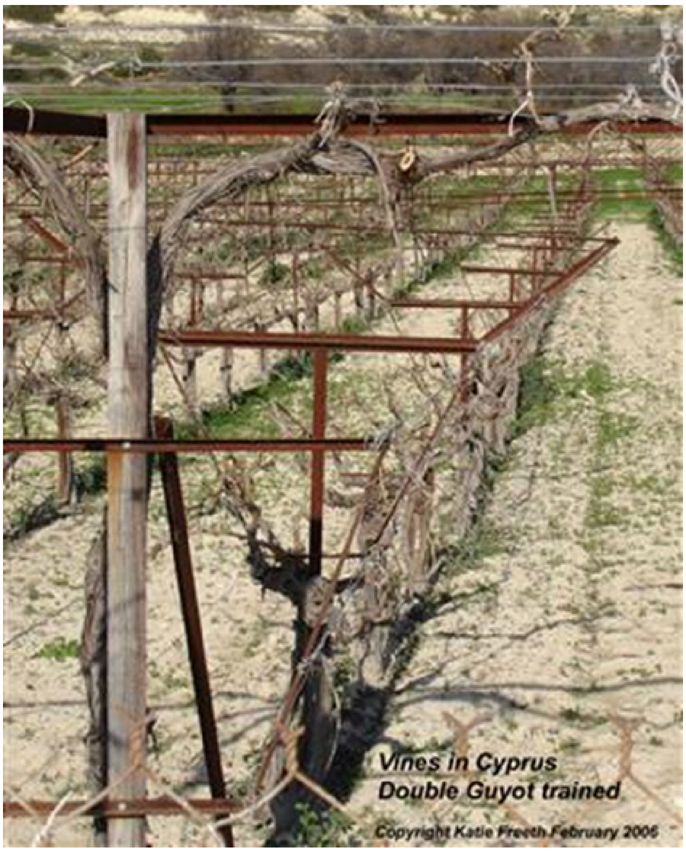
If wooden intermediate posts are used, the wires are stapled to them with galvanised iron staples. If metal such as star droppers are used, the wires are attached with tie-wire or commercial staples made for that purpose.

Guyot system

A form of post and beam wire support is needed for this system. Stout 6-8cm (2-3in) treated posts 2m (6½ ft) long are required, and these should be driven into the ground 60cm/24in deep. Galvanised support wires are secured to the posts, one at 35cm (14in) and at least two double wires at 75cm (30in) and 120cm (4ft). Vines should be planted 1.5m (5ft) apart with at least 1.3m (4ft) between rows.



Guyot system post and wire framework



Double Guyot System

Geneva Double Curtain (GDC) system

Use very strong posts (such as railroad sleepers or telegraph poles). These need to be 3m (10ft) long and are driven 1m (39in) into the ground at 10m (32½ft) intervals. At a height of 1.7m (5½ft) above the ground surface, crossbars 1.2m (4ft) wide are nailed to them. Two galvanised wires are fixed to either side of the crossbars. The vines are spaced at 3m (10ft) intervals.

Four-arm Kniffen system

10 x 10cm (4 x 4in) poles of 2.5m (8ft) long are driven 60cm (2ft) into the ground at spacings of 6.5-7m/ (21-22ft). Galvanised wire is fixed to the posts at heights of 1m (39in) and 1.8m (5¾ ft) above the soil level. Vines are spaced about 2.5m (8ft) apart.

Training Systems

Once a suitable framework for a training system has been put into place, vine growth needs to be controlled via pruning – that is "trained". Although many training systems have been used and discarded through the centuries of grape production, almost all can be grouped into two basic methods: head training and cordoning.

Head Training

In head training, the length of the main stem depends upon the type of grape being grown. In general, this stem is allowed to grow to 1.2-2m (4-6½ feet) high, and is tied to a vertical stake. No more than six laterals are allowed to develop from the top half of the main stem; these stems are chosen so that they are fairly evenly distributed around the stem in a pattern resembling a circle. For the first few winters, these laterals are simply spur-pruned so that a series of short "arms" or branches develops. Fruiting laterals will grow from these branches. Subsequently, as it matures, depending on the type of grape, the vine will be spur-pruned or rod-pruned. In the case of spur-pruned vines, only the vertical stake is required for support. On rod-pruned vines, the fruiting laterals may be left to hang down freely; in other cases they will be supported on horizontal, and sometimes vertical, sets of wires.

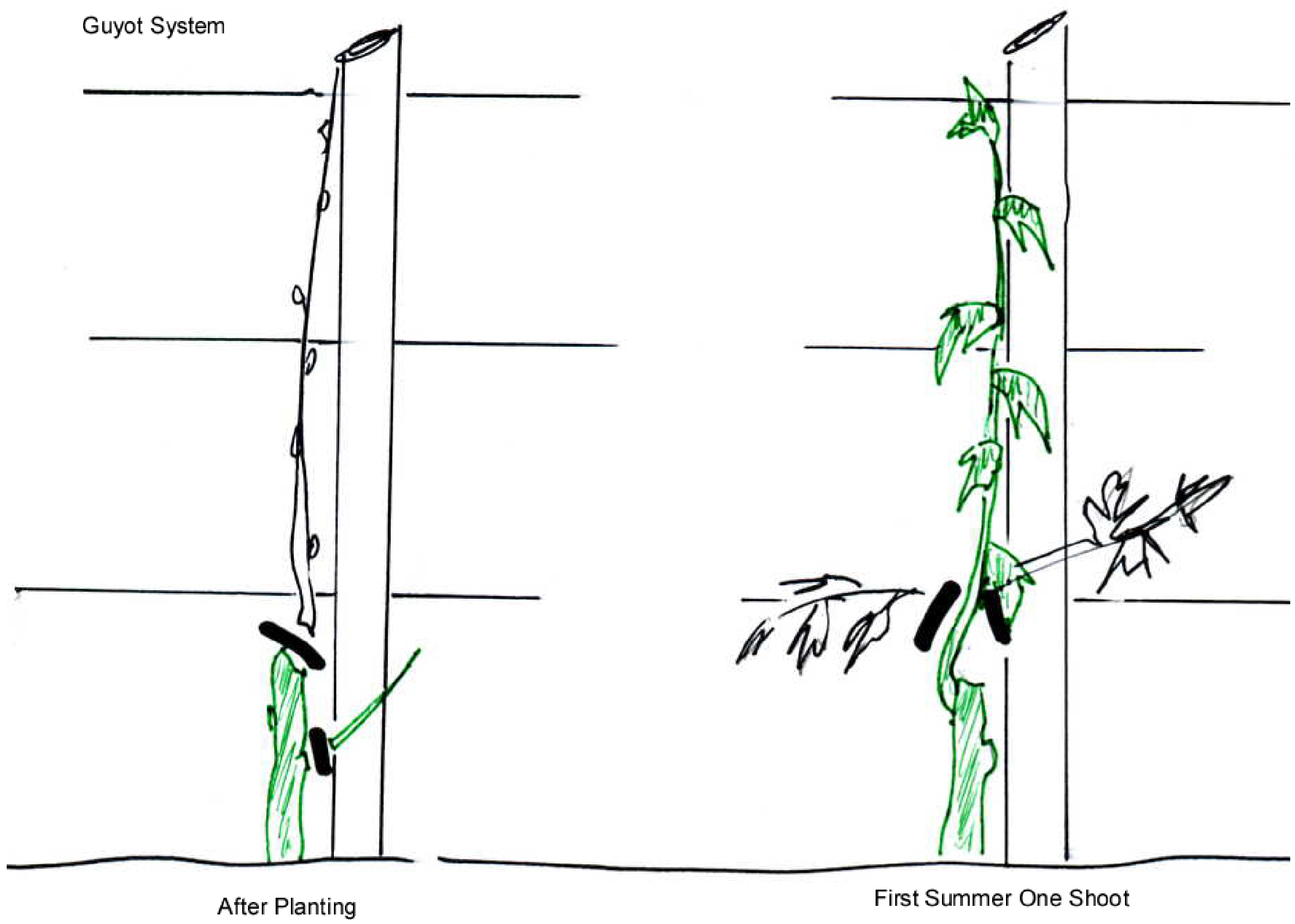
Cordoning

After planting, the main stem is allowed to grow up to a horizontal wire. This stem is then pinched back to just below the wire. Fro the resulting shoots that grow form the stem, only two are selected. These two laterals (the cordons) are then trained along the horizontal wire, one to the right and one to the left. Many of the most common training systems in use are variations of the basic cordon system, e.g. the Guyot system, the Geneva Double Curtain (GDC) and the Kniffen system.

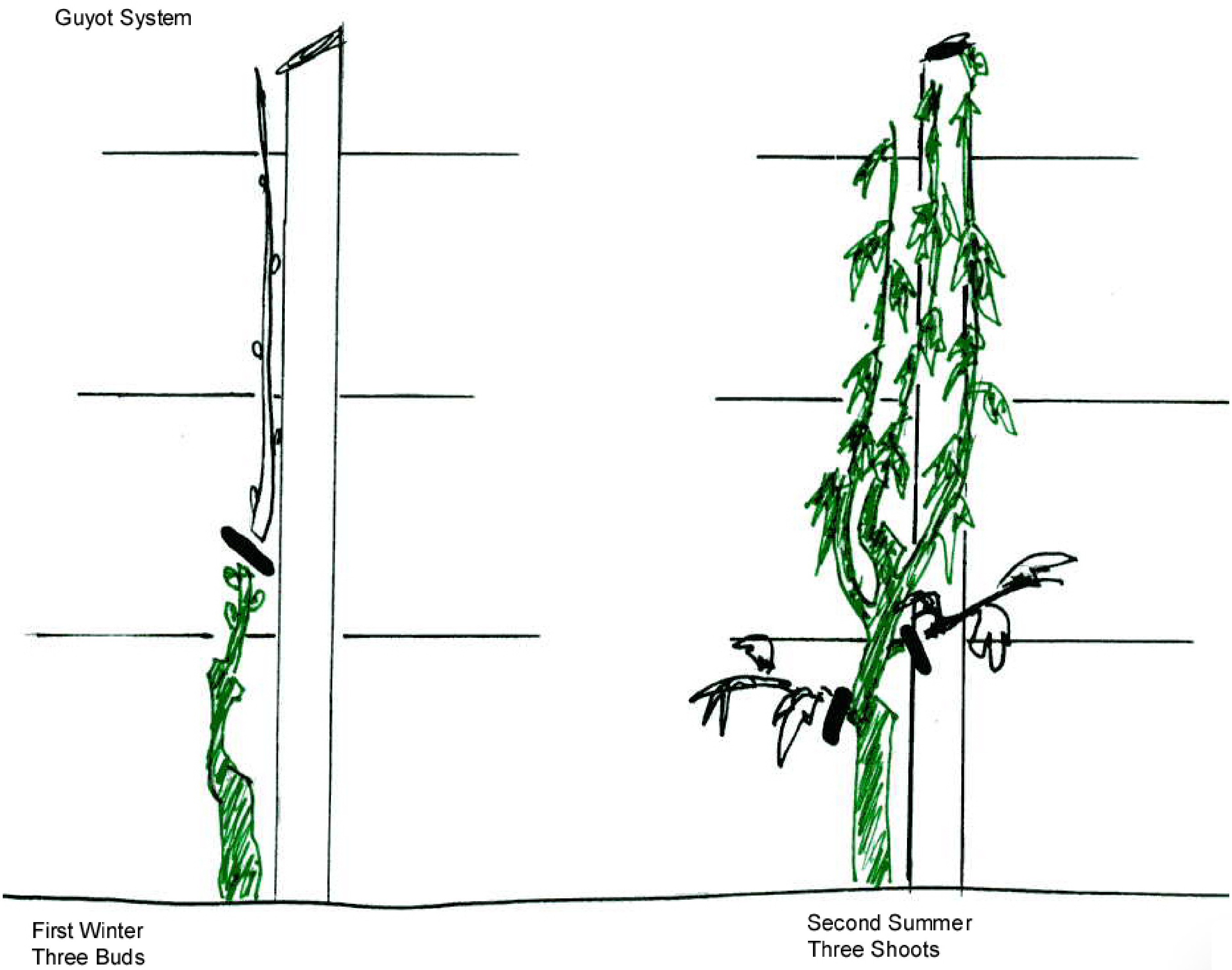
Guyot system

European vines grown in the open are usually trained in the Guyot system, which may be single (a single lateral along the wire in one direction) or more usually double (one lateral along the wire in either direction). The Guyot system is based on annual replacement of the laterals.

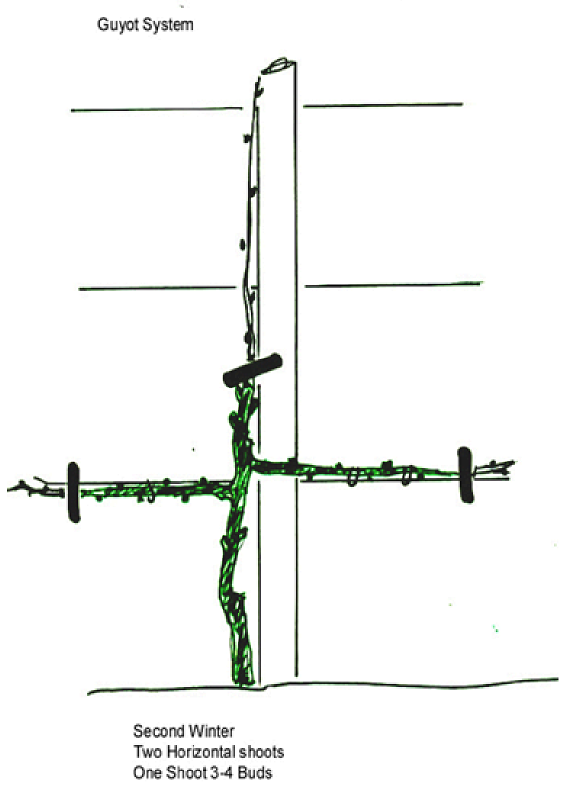
After planting, the vine is cut right down to buds above soil level, or for grafted vines, the graft union. One shoot only is allowed to grow up the vertical support in the first summer. This is cut to 40cm (16in) above the soil level (or graft union) in the first winter, to leave 3 vigorous buds. Only these three buds will be allowed to produce the shoots that grow up the vertical support in the following summer. In the second winter, two of these are tied in to the bottom horizontal wire, one in each direction away from the main stem; they are cut back to 60cm (24 in). The third shoot is cut back to 3 or 4 buds.

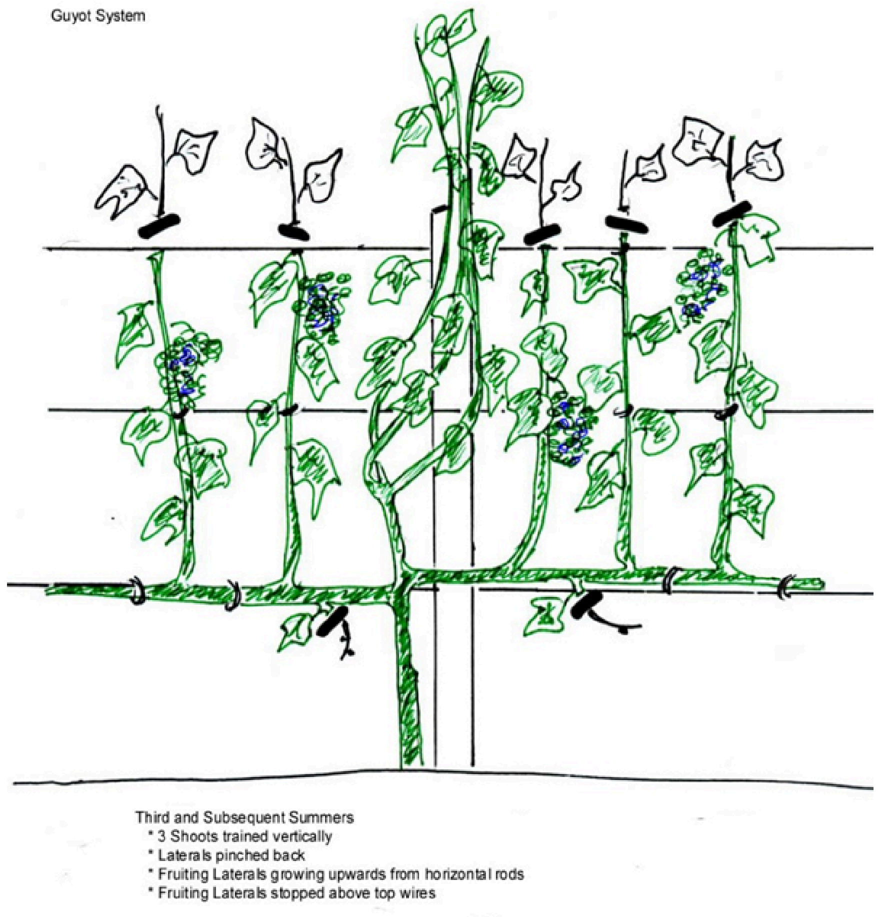


Guyot system – at planting and in the first summer



Guyot system – first winter and second summer





Guyot system – second winter and subsequent summers

From then on during the summer, the three shoots are trained to about 1.75m (6ft) high. Side shoots are controlled by pinching back to one leaf. For optimum production remove flower trusses. From these, fruiting laterals will develop and are trained up through the two sets of double wires. Laterals are not allowed to grow any more than two leaves above the top wire. Towards the end of the summer, foliage which is covering the fruit should be removed, both to encourage ripening and help to prevent diseases such as botrytis.

Guyot system – third and subsequent winters

Post-harvest the laterals are rod-pruned. Three replacement shoots are trained up the support, keeping side shoots pinched back to one leaf. New fruiting laterals will develop from the replacement shoots and the cycle will repeat itself.

Geneva Double Curtain (GDC)

This is a method favoured and perfected in USA; French hybrid vines are often trained this way. There are both advantages and disadvantages to this method, which is best used for really vigorous vines because they must grow high up.

Advantages of GDC System	Disadvantages of GDC System
Machinery and herbicides easier to use	More land is required since yields per hectare are lower than Guyot System
No summer pruning	Vines often take a year (or more) longer to start bearing fruit because of the time spent growing the vines higher
No bending over to harvest	In cooler climates ripening may be delayed

For the Geneva Double Curtain system, after planting the vine is shortened to two buds above the soil level (or for grafted vines, the graft union). Only the more vigorous shoot is allowed to grow up a vertical support – all other shoots are pinched off. Pruning in the first winter depends on how long the shoot has grown:

- For growth of 2m (6½ ft) or more the shoot is cut back to 1m (39in) high
- For growth of 1.75m (5¾ ft) or less the shoot is cut back to 0.5m (19in)

During the second summer, select two main stems growing from the original shoot and tie these, horizontally, to the top wire so that fruiting laterals can grow from them. Cut these back, so that they do not get too long, too quickly. All other shoots should be removed. Some vines are less vigorous and for these, only one horizontal shoot should be tied on in each year. The “double curtaining” effect is achieved by tying in horizontally, the shoots of alternate vines to the lower wire.

The fruiting laterals are selected from shoots growing from the *undersides* of the main laterals, at intervals of approximately 20cm (8in), and are pruned to 3-4 buds. All other fruiting laterals are removed. Once harvesting is finished, the fruiting laterals are spur-pruned to 3-4 buds.

Vines grown as large as this need robust supports, as they are very heavy. Posts are larger and driven deeper into the ground than those for the Guyot system, for example (dimensions are given in the ‘Materials for Construction’ section above).

Four-arm Kniffen System

This system uses two sets of main laterals on each vine; these are tied onto horizontal wires at heights of 0.6m (2ft) and 1.5-2m (4-6ft). For the first year pruning and training is as for the Guyot system, but in the second year, the main stem is grown all the way to the top wire. Growths which develop above the top wire or below the bottom wire are removed. Laterals between the wires are allowed to grow.

In the second winter, two strong laterals close to the bottom wire and two close to the top wire are tied in to the wires and shortened to 10-15 buds each. In the following year a further 4 laterals are pruned to 2 buds each. These will form replacement laterals the following year. All other growth is removed. The fruiting laterals growing from these main laterals hang down freely, without support. Winter pruning is rod-pruning of the main laterals; these are replaced with new laterals growing from the main stem near the wires. Replacement laterals are shortened depending on the cultivar, but usually to 6-10 buds. Other growths are removed.

This system is popular for training American grapes.

Six-arm Kniffen System

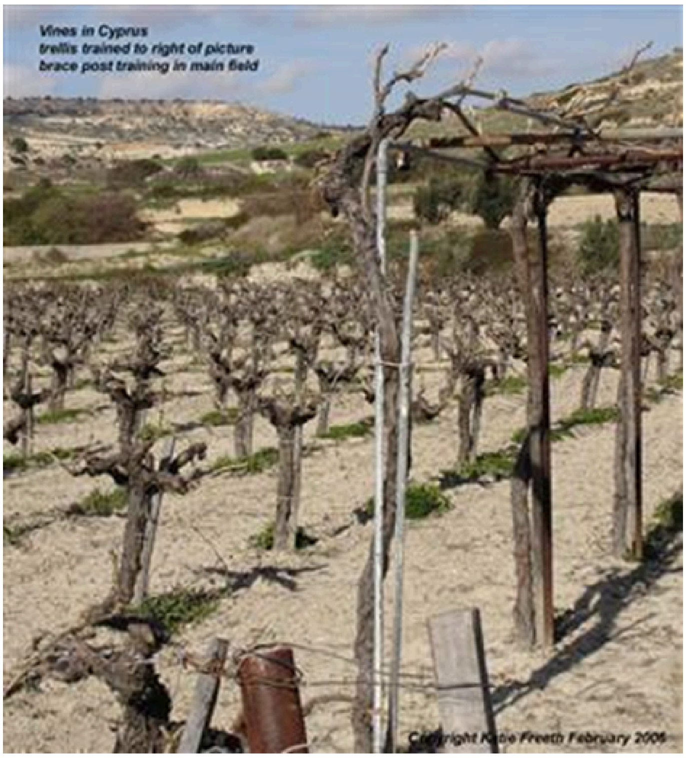
This differs from the four-arm Kniffen system; three horizontal wire supports are used with one pair of laterals at each of the wires. The laterals are usually pruned to only four buds on each. Vines frequently are grown with two main stems. This upright type of system is favoured for French hybrids and Muscadine grapes which have upright growth habits.

Umbrella Kniffen System

This system is popular for American vines. For the first year, pruning and training follow the four-arm Kniffen system. However, in year two, the main stem is pruned to a height of 15-30cm (6-12in) below the top wire. Select three to six fruiting laterals, near the top of the stem and shorten to 6-10 buds. These laterals are allowed to fall over the wire and hang down without support. Two additional laterals are selected to form the replacement laterals for the following year and shortened to 2 buds.

Pergola Training System

The Pergola system is really another Kniffen system; the vine is trained up one side of the pergola, horizontally across the top and down the other vertical side. The number of cross members of the pergola is used to determine the number of laterals on the vine. The fruit grown on pergola trained vines is subject to greater shade than other systems and yields are reduced. This system is popular in Southern Europe and the Mediterranean, particularly in domestic situations, where the vine provides shade over an outside terrace or sitting area.



Vines trained on a high trellis

Vine trained on Wall